

Kavindu Munugoda (PhD)

Pharmacognosy | Phytochemistry | Molecular Virology | Natural Product Drug Discovery

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 LinkedIn  Google Scholar  ORCID iD  ResearchGate  GitHub

Summary

Postdoctoral researcher specializing in natural product drug discovery for cancer and viral diseases, with proven expertise bridging traditional ethnobotanical knowledge and modern molecular approaches. Developed WHO-compliant vector control solutions and characterized 127+ bioactive plant species for therapeutic applications. Current research focuses on HPV oncoprotein-targeting strategies using computational screening and natural product libraries, demonstrating measurable success in translating traditional medicine into evidence-based therapeutics for cancer treatment.

Education

PhD

Pharmacognosy and Phytochemistry

University of Sri Jayewardenepura, Gangodawila, Nugegoda

June 2019- July 2023

Dissertation: Production of Mosquito Repellents Using Lesser Known, Under-Utilized Volatile Oil Producing Plant Species

- Identified 127 novel aromatic plants for mosquito-repellent properties through systematic phytochemical screening
- Developed 5 market-ready mosquito repellent formulations with field-tested efficacy
- Conducted GC-MS analysis, bioactivity screening, and comprehensive field efficacy testing
- Led interdisciplinary research combining natural product chemistry with sustainable biotechnology
- Advanced ethnobotanical knowledge documentation and preservation through scientific research methods
- Conducted multi-site field efficacy trials following WHO vector control guidelines and IAEA standards
- Led community-based research for ethnobotanical knowledge documentation and product acceptance studies
- Implemented WHO pesticide evaluation schemes and international vector control protocols for product development and safety assessment

B.Sc. (Hons) in Applied Sciences

Plant Biotechnology (Major)

2nd Class

University of Sri Jayewardenepura, Nugegoda, Sri Lanka

January 2015 - December 2018

Minored in Chemistry and Zoology

Dissertation: Evaluation of Microbial Counts in Apple Flavoured Beer Compared to Base Beer after Exposing to an External Environment

- Conducted microbiological analysis of apple-flavored beer products over six consecutive days
- Performed microbial identification, total cell counts, and viable cell counts using standard plating medium
- Identified beer spoilage organisms including wild yeasts, *Brettanomyces*, *Lactobacillus*, and *Pediococcus* species
- Demonstrated increased microbial susceptibility in flavored beer due to altered pH and ferment conditions
- Applied standard microbiological testing protocols for fermented beverage quality control

Current Research Position

Visiting Scientist

International Centre for Genetic Engineering and Biotechnology (ICGEB), China Regional Research Centre, Taizhou City, Jiangsu Province, China

December 2024 - Present

Current Project: Ethnobotanical-driven drug discovery targeting HPV oncoprotein-host interactions

- Screening bioactive small molecules from Sri Lankan medicinal plants
- HPV11 E1/E2 protein cloning, expression, and functional characterization
- HPV16 E6/E7 therapeutic target validation using natural compounds
- **Cell lines utilized:** HEK293T, HaCaT, HeLa, CaSki, C33A
- **Techniques:** Protein-DNA interaction assays, luciferase reporter systems, Western blot analysis

Research Assistant

Molecular Medicine Unit, Faculty of Medicine, University of Kelaniya, Sri Lanka

December 2023 - November 2024

- Advanced molecular analysis protocols and hatchery management systems
- Implementation of cutting-edge molecular medicine techniques for therapeutic research
- Collaborative interdisciplinary research in molecular medicine and drug development
- Mentored undergraduate students in advanced molecular techniques

Patents

Volatile Plant Oil–Based Mosquito Repellent Sticks & Sprays

Application LK/P/1/23236

National Intellectual Property Office, Sri Lanka

July 2024

Co-inventors: S. M. C. U. P. Subasinghe; K. D. Munugoda; M. D. Hapugoda

Publications

Published in Peer-Reviewed Journals

- **Munugoda, K. D.**, Thalagala, T.S., Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2023). Choice modelling for the commercial cultivation of underutilized aromatic plants for producing mosquito repellents: Targeting rural sector income generation. *Economic Botany* (2023), 1-19. <http://dx.doi.org/10.1007/s12231-023-09573-y>
- Chen, J., Lee, S. Y., **Munugoda, K. D.**, Mohamed, R., Subasinghe, S. M. C. U. P., & Liao, W. (2021). The complete chloroplast genome of Walla Patta, *Gyrinops walla* (Thymelaeaceae), an agarwood-producing tree species from Sri Lanka. *Mitochondrial DNA Part B*, 6(6), 1699-1701. <http://dx.doi.org/10.1080/23802359.2021.1926362>

Submitted Manuscripts

- **Munugoda, K. D.**, Wijesekara, W. L. I., Hettiarachchi, D. S., Rajapaksha, S. M., Subasinghe, S. M. C. U. P., Cooray, A. T., & Jayanika, R. G. I. (2025). Reinventing Artisanal Curry Leaf Paste: Biochemical and Sensory Analysis of Five Sri Lankan Cultivars. *agRxiv*. doi:10.31220/agRxiv.2025.00356 (Preprint)
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2024) Refining Nature's Defense: Production and Evaluation of Mosquito Repellents from *Ocimum tenuiflorum* L. (Krishna and Rama Subtypes) and *Ocimum gratissimum* L. Essential Oils doi:10.26434/chemrxiv-2025-5l0v3(Preprint)
- **Munugoda, K. D.**, Wijesekara, W.L.I., Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2023). Reinventing Artisanal Curry Leaf Paste: Biochemical and Sensory Analysis of Five Distinct Sri Lankan Cultivars (under revision)

Presentations

Oral Presentations

- Perera M.S.S., Fernando P.J.S., De Silva P.R.S., and **Munugoda K.D.** (2022). Willingness to implement a hydroponics system among Up-country vegetable farmers in Sri Lanka. International Open University Research Sessions (iOURS 2022), Open University, Sri Lanka
- Perera M.S.S., Fernando P.J.S., De Silva P.R.S., and **Munugoda K.D.** (2022). Farmers' Perceptions and Willingness to Adopt Hydroponic Systems: A Study in Welimada, Sri Lanka. International Symposium on Agriculture and Environment (ISAE 2022), Faculty of Agriculture, University of Ruhuna, Sri Lanka
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2022). Medicinal aromatic plant cultivation for the bioeconomy: Eliciting willingness from rural communities. Environmental Economics and Resource Management, 26th International Forestry and Environment Symposium 2021, University of Sri Jayewardenepura, Sri Lanka
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2020). Potential of extra income generation for the rural Sector by medicinal plant cultivation. Impacts of COVID-19 on Society and Environment, 25th International Forestry and Environment Symposium 2020, University of Sri Jayewardenepura, Sri Lanka

Poster Presentations

- Munugoda, K. D., Buddhinie, P. K. C., & De Silva, C. (2019). Evaluation of microbial counts in apple flavoured beer compared to base beer after exposing to external environment. Uni-In Alliance 2019, University of Sri Jayewardenepura, Sri Lanka

Grants

ICGEB-DIC-MOST International Fellowship
December 2024

Awards

Best Oral Presenter

26th International Forestry Symposium
January 2022

Session: Environmental Economics and Resource Management

Title: Potential of Extra Income Generation for the Rural Sector by Medicinal Plant Cultivation

Certifications

PacMOSSI-Aedes & Anopheles Surveillance & Control
James Cook University, Australia
June 2024

Thermo Fisher Scientific - Mass Spectrometry Job Simulation
Forage
January 2024

Research Methodology and Scientific Writing
Faculty of Graduate Studies, University of Sri Jayewardenepura, Sri Lanka
November 2020

Science, Technology and Innovation Policy and Policy Instruments for Sustainable Development Goals for Asia and the Pacific
UN Interagency Task Team on STI for the SDGs (IATT)
November 2024

Guest Lectures

Department of Agricultural and Food Technology
University of Vocational Technology, Dehiwala-Mount Lavinia , Sri Lanka
2019-2021

Research methodology (undergraduates)

Department of Forestry and Environmental Sciences
University of Sri Jayewardenepura, Gangodawila, Nugegoda, Sri Lanka
2019-2023

Course: Research methodology (undergraduates and master students)

Memberships

Associate (A. I. Chem.C.)

Membership No: 3548 Aff

The Institute of Chemistry Ceylon

Affiliated to Federation of Asian Chemical Societies (FACS), International Union of Pure and Applied Chemistry (IUPAC), Royal Society of Chemistry, UK and Royal Australian Chemical Institute.

<https://ichemc.ac.lk/>

Languages

English

IELTS 7.0



References

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